

Group 1

Inputs:

1. **Team Work Plan** : - members name missing
 2. Split L1 Cache Hierarchy and Dynamic Branch Prediction, is this sufficient as in our case memory latency is one cycle, do we need cache?.
 3. **Clarify the branch-prediction claim:** A **0-cycle branch penalty** is difficult in a conventional 5-stage pipeline unless prediction and target generation are handled carefully in the Fetch stage. The design should clearly explain how this is achieved
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Group 2

Inputs:

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
 2. **Timeline planning:** Many activities are scheduled for the last 2 weeks, please move activities to earlier weeks. (design goals are not mentioned clearly in timeline)
 3. Design goals must be quantitative and trackable - please include the value of latency, cycle and others.
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Group 3

Inputs:

1. **Detailed Timeline:** Given the scope, a more detailed timeline plan in the next report submission will help. Identify the specific modules (hardware in Verilog and software in Python/C/other languages) required for the demo. For each module, clearly specify: (i) design completion, (ii) testing/verification with testbench, and (iii) readiness for integration/demo, along with expected deliverables. Once this is defined, re-check the owners and verify timeline feasibility.
 2. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited.
 3. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
 4. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
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Group 4

Inputs:

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited.
 2. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
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Group 5

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited.
 2. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
 3. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 4. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle
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Group 6

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
 2. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 3. **Design Goals:** Design Goals should be quantitative and trackable.
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Group 7

1. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle.
 2. Design goal should be quantitative and trackable
 3. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
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Group 8

1. Title says 5 Stage pipelined but the description says 3 Stage pipeline
2. Given the AI tool support (example: <https://chatgpt.com/share/69c8b008-4f60-8320-a9d0-58af100f4fc1>), implementing only basic features like MUL and DIV (or MUL-ADD) would likely receive a low score (~9/30).

Even if your current plan is slightly beyond this, it still gives a rough idea of the expected marks.

Perhaps, you can consider expanding the project scope, especially by increasing the hardware implementation—either in the minimum system or the project goals.

Group 9

1. What are the limits on the input sizes considering the hardware resources are limited ?
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Group 10

Inputs :

1. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 2. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
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Group 11

Inputs :

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
 2. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
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Group 12

Inputs :

1. Inputs sizes: What are the limits on the input sizes considering the hardware resources are limited
 2. Resource Estimate: Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 3. Cache Requirement: What is the need of L1 cache and data cache when current latency is 1 cycle.
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Group 13

Inputs :

1. Inputs sizes: What are the limits on the input sizes considering the hardware resources are limited
 2. Cache Requirement: What is the need of L1 cache and data cache when current latency is 1 cycle.
 3. Please consider expanding the project scope, especially by increasing the hardware implementation
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Group 14

Inputs:

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
 2. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 3. **Scheduling Order:** Order of implementation should be Commitment first then goal.
 4. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle.
 5. **Design Goals:** Design Goals should be quantitative and trackable
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Group 15

Inputs:

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited
 2. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 3. **TimeLine:** Tasks can be done parallelly you are doing serially.
 4. **Order:** Order of implementation should be Commitment first then goal.
 5. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle.
 6. **Design Goals:** Design Goals should be quantitative and trackable
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Group 16

Inputs:

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited.
 2. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
 3. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
 4. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle
 5. **Detailed Timeline:** Given the scope, a more detailed timeline plan in the next report submission will help. Identify the specific modules (hardware in Verilog and software in Python/C/other languages) required for the demo. For each module, clearly specify: (i) design completion, (ii) testing/verification with testbench, and (iii) readiness for integration/demo, along with expected deliverables. Once this is defined, re-check the owners and verify timeline feasibility.
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Group 17

Inputs:

1. Time Complexity for the Algorithm
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Group 18

1. **Prioritize the minimum commitment:** The project includes RV32M, filtering, UART upload, multiple filters, performance comparison, and possible pipelining. Ensure the **fixed 64×64 Gaussian accelerator and FPGA demonstration** are completed first before focusing on stretch goals.
 2. **Integration risk should be managed early:** Since the existing 3-stage processor is being modified while the accelerator is developed in parallel, establish the **memory-mapped register interface and BRAM ownership early**. Weekly integration with small end-to-end tests will reduce the risk of last-minute processor–accelerator integration problems.
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Group 19

Inputs:

1. **Add more comprehensive performance comparison:** The project already includes latency and throughput counters, but comparing the FPGA implementation with a **software-based equivalent** would make the acceleration benefit much stronger. This is currently listed as a stretch goal and could be a valuable final evaluation if time permits.
 2. **Latency target needs careful validation:** The ≤ 5 -cycle decision latency and 100 MHz target are good measurable goals, but the team should clearly define whether latency includes packet parsing, packet acceptance, and decision generation to make the measurement consistent.
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Group 20

Inputs:

1. Inputs sizes: What are the limits on the input sizes considering the hardware resources are limited
 2. Resource Estimate: Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
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Group 21

1. **Input sizes:** The proposal mentions a 640×480 VGA display resolution and host PC streaming over UART, but it does not specify strict input size limits for incoming image .
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Group 22

1. **Inputs sizes:** What are the limits on the input sizes considering the hardware resources are limited.
2. **Exponential Complexity (Algo):** Are there cases where the algorithm has exponential complexity. If so, how is it handled?
3. **Resource Estimate:** Please estimate the LUT, FF, BRAM, and DSP usage, along with percentage utilization. Also, will the design fit on the target FPGA.
4. **Cache Requirement:** What is the need of L1 cache and data cache when current latency is 1 cycle
5. **Scope:** Given the AI tool support (example: <https://chatgpt.com/share/69c8b008-4f60-8320-a9d0-58af100f4fc1>), implementing only basic features like MUL and DIV (or MUL-ADD) would likely receive a low score (~9/30).

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Group 23

1. **Scope:** Given the AI tool support (example: <https://chatgpt.com/share/69c8b008-4f60-8320-a9d0-58af100f4fc1>), implementing only basic features like MUL and DIV (or MUL-ADD) would likely receive a low score (~9/30).

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